

$\theta_{\text{ATE\%}}$ considered in [Section 4.1](#).⁵³

We note, however, that if one were instead to impose (C.4) with the assumption that $E[\log(U(d)) | D] = 0$, then the regression of $\log(Y)$ on D would be consistent for β , whereas Poisson regression would generally be inconsistent for β . Indeed, under the potential outcomes model in (C.5) with the assumption that $E[\log(U(d))] = 0$, we have that $\beta = E[\log(Y(1)) - \log(Y(0))]$, the ATE in logs.⁵⁴

This discussion highlights that whether or not an estimator is consistent depends on the specification of the *target parameter*. Our results help to illuminate what parameters can be consistently estimated by enumerating the properties that identified causal parameters can (or cannot) have.

C.3 Tobit models

An alternative structural approach is to explicitly model the extensive margin, a classic example of which is the Tobit model ([Tobin, 1958](#)). Following the discussion of Tobit models in [Angrist and Pischke \(2009\)](#), suppose there exist latent potential outcomes $Y^*(d) = \mu_d + U$, where $U \sim \mathcal{N}(0, \sigma^2)$ and $D \perp U$. The observed potential outcome $Y(d)$ is then the latent potential outcome truncated at zero, $Y(d) = \max(Y^*(d), 0)$. We note that in this model, the treatment has a constant additive effect of $\mu_1 - \mu_0$ on the latent outcome, and the latent potential outcomes are assumed to be normally distributed.

Thanks to the parametric assumptions, the unknown parameters μ_1, μ_0, σ^2 are identified and estimable via, e.g., maximum likelihood. As a result, the entire joint distribution of potential outcomes is identified, since this depends only on (μ_1, μ_0, σ) . This implies, in turn, that all of the possible target parameters considered in [Section 4](#) are point-identified. For example, under this model

$$E[\log Y(d) | Y(1) > 0, Y(0) > 0] = E[\log(\mu_d + U) | U > -\mu_1, U > -\mu_0],$$

where the right-hand side can be computed numerically since $U \sim \mathcal{N}(0, \sigma^2)$. Thus, the intensive margin treatment effect in logs, $\theta_{\text{Intensive}}$, is actually point-identified under the Tobit model.⁵⁵

It is worth noting that unlike some of the models considered above, the Tobit model is consistent with a nonzero extensive margin. However, the assumptions of normal errors and constant treatment effects on the latent index are restrictive. As discussed in [Section 4](#), imposing these assumptions is not necessary for identification if one is ultimately interested in, say, $E[Y(1) - Y(0)]/E[Y(0)]$, and one can obtain bounds on the intensive margin effect without imposing these assumptions.⁵⁶ Moreover, as [Angrist and Pischke \(2009\)](#) and [Angrist \(2001\)](#) point out, it is often not clear what the economic meaning of the latent potential outcome $Y^*(d)$ is—if $Y(d)$ is earnings, for example,

⁵³[Bellégo et al. \(2022\)](#) also consider (C.4), but consider the more general class of identifying restrictions of the form $E[D \log(U + \delta)] = 0$, where δ is a tuning parameter.

⁵⁴Note that the assumption that $E[\log(U)] = 0$ implicitly implies that $U > 0$, and thus $Y > 0$.

⁵⁵Likewise, the intensive margin treatment effect in levels, $E[Y(1) - Y(0) | Y(1) > 0, Y(0) > 0]$ is simply $\mu_1 - \mu_0$.

⁵⁶We note that the assumptions of the Tobit model imply (but are strictly stronger than) the assumption of rank preservation of the potential outcomes. However, rank preservation alone suffices to point identify $E[\log Y(1) - \log Y(0) | Y(1) > 0, Y(0) > 0]$.

what is the meaning of having negative latent earnings ($Y^*(d) < 0$)?

D Connection to two-part models

One approach recommended for settings with weakly-positive outcomes is to estimate a two-part model (Mullahy and Norton, 2023). In this section, we briefly review two-part models, and show that the marginal effects implied by these models do not correspond with ATEs for the intensive margin without further restrictions on the potential outcomes. Thus, while two-part models strike us as a reasonable approach if the goal is to model the conditional expectation function of observed outcomes Y given treatment D (as in Mullahy and Norton (2023)), they will often not be appropriate if instead the goal is to learn about a causal effect along the intensive margin.⁵⁷

The idea of a two-part model is to separately model the conditional distribution $Y \mid D$ using (a) a first model for the probability that Y is positive given D , $P(Y > 0 \mid D)$ (b) a second model for the conditional expectation of Y given that it is positive, $E[Y \mid D, Y > 0]$. Common specifications include logit or probit for part (a), and a linear regression of the positive values of Y on D for part b); see, e.g., Belotti et al. (2015). After obtaining estimates of the two-part model, it is common to evaluate the marginal effects of D on both parts, i.e. the implied values of

$$\begin{aligned}\tau_a &= P(Y > 0 \mid D = 1) - P(Y > 0 \mid D = 0) \\ \tau_b &= E[Y \mid Y > 0, D = 1] - E[Y \mid Y > 0, D = 0].\end{aligned}$$

We now consider how the parameters of the two-part model relate to causal effects in the potential outcomes model. Suppose, for simplicity, that the two-part model is well-specified, so that it correctly models $P(Y > 0 \mid D)$ and $E[Y \mid Y > 0, D]$. Suppose further that D is randomly assigned, $D \perp\!\!\!\perp Y(1), Y(0)$. In this case, we have that

$$\begin{aligned}\tau_a &= P(Y(1) > 0) - P(Y(0) > 0) \\ \tau_b &= E[Y(1) \mid Y(1) > 0] - E[Y(0) \mid Y(0) > 0].\end{aligned}$$

From the previous display, we see that the marginal effect on the first margin, τ_a , has a causal interpretation: it is the treatment’s effect on the probability that the outcome is positive.

The interpretation of the marginal effect on the second margin, τ_b , is more complicated, however. For simplicity, suppose we are willing to impose the “monotonicity” assumption discussed in Section 4, $P(Y(1) = 0, Y(0) > 0) = 0$, so that anyone with a zero outcome under treatment also has a zero outcome under control. Then, letting $\alpha = P(Y(0) = 0 \mid Y(1) > 0)$, we can write τ_b as

$$\begin{aligned}\tau_b &= (1 - \alpha)E[Y(1) \mid Y(1) > 0, Y(0) > 0] \\ &\quad + \alpha E[Y(1) \mid Y(1) > 0, Y(0) = 0] - E[Y(0) \mid Y(1) > 0, Y(0) > 0]\end{aligned}$$

⁵⁷We are particularly grateful to John Mullahy for an enlightening discussion of this topic.

$$\begin{aligned}
&= \underbrace{E[Y(1) - Y(0) \mid Y(1) > 0, Y(0) > 0]}_{\text{Intensive margin effect}} \\
&\quad + \alpha \underbrace{(E[Y(1) \mid Y(1) > 0, Y(0) = 0] - E[Y(1) \mid Y(1) > 0, Y(0) > 0])}_{\text{Selection term}},
\end{aligned}$$

where the first equality uses iterated expectations, and the second re-arranges terms.

The previous display shows that τ_b is the sum of two terms. The first is the ATE for individuals who would have a positive outcome regardless of treatment status (similar to $\theta_{\text{Intensive}}$ in [Section 4](#), except using Y instead of $\log(Y)$). The second term is not a causal effect, but rather represents a selection term: it is proportional to the difference in the average value of $Y(1)$ for “compliers” who would have positive outcomes only under treatment versus “always-takers” who would have positive outcomes regardless of treatment status. In many economic contexts, we may expect this selection effect to be negative. For example, we may suspect that individuals who would only get a job if they receive a particular training have lower ability, and hence lower values of $Y(1)$, than individuals who would have a job regardless of training status. The marginal effect τ_b thus only has an interpretation as an ATE along the intensive margin if either (a) there is no extensive margin effect ($\alpha = 0$) or (b) we are willing to assume that the selection term is zero. [Angrist \(2001\)](#) provided a similar decomposition (without imposing monotonicity), concluding that the two-part model “seems ill-suited for causal inference,” at least without further restrictions on the potential outcomes. See, also, [Mullahy \(2001\)](#) for additional discussion.

E Details on Lee bounds using IV in [Berkouwer and Dean \(2022\)](#)

We now describe in detail our approach for constructing [Lee \(2009\)](#)-type bounds in the IV setting of [Berkouwer and Dean \(2022\)](#).

Estimating the instrument-complier distributions. The first step is to estimate the distribution of $Y(0)$ and $Y(1)$ for instrument-compliers. As shown in [Abadie \(2002\)](#), the CDF for $Y(1)$ for instrument-compliers at a point y can be consistently estimated by using two-stage least squares to estimate the effect of treatment on the outcome $D_i 1[Y_i \leq y]$. The CDF for $Y(0)$ for instrument-compliers can analogously be obtained using the outcome $(D_i - 1) 1[Y_i \leq y]$. We estimate these TSLS regressions using analogues to (11) (except replacing $\text{arcsinh}(Y_i)$ with the outcomes just described) for all values of y contained in the data. We thus obtain empirical estimates of the CDFs for instrument-compliers, $\hat{F}_{Y(d)}(y)$ for $d = 0, 1$.

Constructing bounds. Note that if $U \sim U[0, 1]$, then $Y(d) \sim F_{Y(d)}^{-1}(U)$, where $F_{Y(d)}^{-1}(u) := \inf\{y \mid F_{Y(d)}(y) \geq u\}$. With this formulation in mind, [Lee \(2009\)](#)’s bounds for $E[\log(Y(1)) \mid Y(1) > 0, Y(0) > 0]$ can be written as

$$E[\log(F_{Y(1)}^{-1}(U)) \mid U \in [\theta_{NT}, \theta_{NT} + \theta_{AT}]] \leq E[\log(Y(1)) \mid Y(1) > 0, Y(0) > 0]$$

$$\leq E[\log(F_{Y(1)}^{-1}(U)) \mid U \in [1 - \theta_{AT}, 1]], \quad (\text{E.1})$$

where $\theta_{AT} = P(Y(1) > 0, Y(0) > 0)$, $\theta_{NT} = P(Y(1) = 0, Y(0) = 0)$, and $\theta_C = P(Y(1) > 0, Y(0) = 0)$. We estimate the bounds in (E.1) by plugging in the estimated CDFs for instrument-compliers described above, as well as the values of $\theta_{AT}, \theta_{NT}, \theta_C$ implied by the estimated CDFs. We approximate the expectation over U by taking the average over 100,000 uniform draws.⁵⁸ Finally, to compute the bounds on the treatment effect, we must estimate $E[Y(0) \mid Y(0) > 0]$. To do this, we use the fact that

$$E[Y(0) \mid Y(0) > 0] = E[F_{Y(0)}^{-1}(U) \mid U \in [\theta_{NT} + \theta_C, 1]].$$

As before, we then estimate the right-hand-side in the previous display by plugging-in the estimated CDF for instrument-compliers, and simulating over 100,000 uniform draws. The Lee bounds for $\theta_{\text{Intensive}}$ are then obtained by subtracting the estimate of $E[Y(0) \mid Y(0) > 0]$ from the estimates of the lower and upper bounds in (E.1). We estimate standard errors for the bounds using 1,000 draws from a non-parametric clustered bootstrap.⁵⁹

F Appendix Tables and Figures

- Appendix Table 1 contains information on the *AER* papers discussed.
- Appendix Figure 1 shows how t -statistics change in the replication exercise.
- Appendix Table 2 shows the analogue of Table 1 for $\log(1 + Y)$.

⁵⁸We note that in finite samples, the estimated CDF $\hat{F}_{Y(d)}(y)$ may be non-monotonic. Nevertheless, the inverse $\hat{F}_{Y(d)}^{-1}(u) := \inf\{y \mid \hat{F}_{Y(d)}(y) \geq u\}$ remains well defined.

⁵⁹One complication that arises is that for some draws from the bootstrap distribution, the sign of the extensive margin can be the opposite of that in the original data. In our bootstrap procedure, we construct Lee-type bounds assuming monotonicity in whichever direction matches the bootstrapped data. The resulting bootstrap estimates of the bounds appear to be approximately normally distributed, but we think a formal theoretical evaluation of the bootstrap in this setting is an interesting topic for future work.